CHANG LIU

716-907-4211 | changliu.chris@tamu.edu | LinkedIn | Google Scholar | Website | GitHub

EDUCATION

Texas A&M University — College Station, TX
Ph.D., Mechanical Engineering – Advisor: Dr. Minghui Zheng; GPA 3.83/4.0 — *Jan. 2024 – Dec. 2027 (expected)*

University at Buffalo, SUNY — Buffalo, NY
M.S., Mechanical Engineering – Advisor: Dr. Minghui Zheng; GPA 3.83/4.0 — *Aug. 2018 – Feb. 2021*
- Thesis: *Deep Learning-Based Collision Prediction and Motion Planning for Collaborative Robots*.

Huazhong University of Science and Technology — Wuhan, China
B.S., Energy and Power Engineering — *Sep. 2014 – June 2018*

RESEARCH INTERESTS

My research focuses on robot learning for reliable, contact-rich manipulation, particularly learning from human corrections and recovering from failures. I study how vision-language-action policies and structured manipulation skills can work together, using robotic disassembly as a real-world testbed.

RESEARCH EXPERIENCE

Graduate Research Assistant — College Station, TX
Control and Robotics Lab, Texas A&M University – Advisor: Dr. Minghui Zheng — *Jan. 2024 – Present*

Robot Learning for Contact-Rich Disassembly — *Jan. 2024 – Present*
- Developed **PACER** (CoRL 2026), a process-aware reweighting framework that learns from autonomous rollouts, human corrections, and clean demonstrations through **native supervised fine-tuning (SFT)**.
- PACER achieved **87/100 successful approach-and-handoff trials**, versus 54/100 for vanilla post-SFT, on a held-out UR5e benchmark covering five motherboard component classes.
- Developed **SELF-VLA**, coordinating VLA approach and recovery policies with programmed disassembly and placement skills through stop-token handoffs.
- SELF-VLA increased full-task success from **23.7% to 76.0%** over end-to-end $\pi_{0.5}$ across seven component tasks (350 trials per method); OpenVLA-OFT increased from 0.0% to 63.4% (revised manuscript under review).

Robot Learning Software and Datasets — *Jan. 2024 – Present*
- Built and open-sourced a teleoperation and data collection platform on GELLO, integrating UR5e control through ZMQ, dual-camera capture, and phase-labeled recording and export for OpenVLA, OpenVLA-OFT, and openpi training and inference.
- Developed **dataset annotation software** and a separate **dual-joystick teleoperation system**.
- Contributed to the **End-of-Life Desktop Multimodal Dataset**, a team collection of 20,000+ original images across 15+ device models and 500+ complete video demonstrations; preparing the dataset manuscript for CVPR submission.
- **E-Waste Component Segmentation Dataset** comprises 1,456 annotated laptop-component images under varied lighting and viewpoints, used for SAM2 and YOLOv8 benchmarking.

RAISE: Robotic Disassembly and Sorting System — *Jan. 2024 – Present*
- Designed and built the **robotic sorting station** for end-of-life phones with Idaho National Laboratory, integrating UR5e hand-eye calibration, YOLOv8 instance segmentation, and automated sorting. The perception model achieved **98.9% test accuracy at IoU 0.7** on 300 images.
- Integrated the station with cutting and battery-removal subsystems; the published RAISE system reports processing over 120 phones per hour, and its techno-economic analysis projects that the system turns a previously unprofitable process into a net profit per unit weight of phones processed.

Graduate Research Assistant — Buffalo, NY
CAL-UB Lab, University at Buffalo, SUNY – Advisor: Dr. Minghui Zheng — *Aug. 2019 – Feb. 2021*
- Developed a vision-only ConvLSTM model to predict robot and obstacle states five steps ahead and support stop-go collision avoidance without access to robot dynamics.
- Built the complete motion-planning dataset and applied SAC reinforcement learning for safety-aware planning in shared human-robot workspaces.
- Built a transfer-learning pipeline for quadrotor propeller fault detection from flight-audio spectrograms.
- Contributed to task-constrained hybrid motion planning for collaborative robots in intelligent remanufacturing (published in *Mechatronics*).
- Set up, calibrated, and maintained laboratory equipment including a Vicon motion capture system, UR5e manipulators, and quadrotor platforms.

PUBLICATIONS

*My name appears in bold (**C. Liu**). Full list also available on Google Scholar.*

Peer-Reviewed Conference Papers

[1] **C. Liu**, S. Tian, L. Gong, Z. Fan, J. Li, X. Liang, M. Zheng. "PACER: Process-Aware VLA Policy Learning from Human Corrections for Robotic Disassembly." *Conference on Robot Learning (CoRL)*, 2026.

[2] J. Lei, **C. Liu**, Y. She, X. Liang, M. Zheng. "Learning When to See and When to Feel: Adaptive Vision-Torque Fusion for Contact-Aware Manipulation." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, PA, 2026.

[3] **C. Liu**, S. Tian, S. Behdad, X. Liang, M. Zheng. "Vision-Language-Action Models for Selective Robotic Disassembly: A Case Study on Critical Component Extraction from Desktops." *Proceedings of the ASME 2026 21st International Manufacturing Science and Engineering Conference (MSEC2026)*, State College, PA, 2026, Paper No. MSEC2026-181638.

[4] **C. Liu**, W. Liu, Z. Chen, M. Zheng. "A Deep-ConvLSTM Collision Prediction Model for Manipulators in Dynamic Environment." *IFAC-PapersOnLine* 55(27) (2022), 68–75.

Peer-Reviewed Journal Articles

[5] **C. Liu**, S. Behdad, P. Pagilla, X. Liang, M. Zheng. "Transforming Remanufacturing Automation with Large Language Models: A Forward-Looking Analysis with Case Studies." *Robotics and Computer-Integrated Manufacturing* 103 (2027), 103361.

[6] X. Zhang, **C. Liu**, X. Liang, M. Zheng, S. Behdad. "Evaluating Large and Lightweight Vision Models for Irregular Component Segmentation in E-Waste Disassembly." *Journal of Manufacturing Science and Engineering* 148(10) (2026), 101005.

[7] **C. Liu**, B. Balasubramaniam, N. A. Yancey, M. H. Severson, A. Shine, P. Bove, B. Li, X. Liang, M. Zheng. "RAISE: A Robot-Assisted Selective Disassembly and Sorting System for End-of-Life Phones." *Resources, Conservation and Recycling* 225 (2026), 108609.

[8] H. Zhao, **C. Liu**, B. Balasubramaniam, J. Li, J. Song, X. Liang, M. Zheng, B. Li. "Precision 3D Profilometry of Consumer-Grade Computer Enclosures Using High Dynamic Range Fringe Projection." *Applied Optics* 64(30) (2025), 8986–8994.

[9] W. Liu, **C. Liu**, X. Liang, M. Zheng. "A Hybrid Task-Constrained Motion Planning for Collaborative Robots in Intelligent Remanufacturing." *Mechatronics* 102 (2024), 103222.

[10] W. Liu, **C. Liu**, S. Sajedi, H. Su, X. Liang, M. Zheng. "An Audio-Based Risky Flight Detection Framework for Quadrotors." *IET Cyber-Systems and Robotics* 6(1) (2024), e12105.

[11] Z. Chen, **C. Liu**, H. Su, X. Liang, M. Zheng. "A Hybrid Disturbance Observer for Delivery Drone's Oscillation Suppression." *Mechatronics* 88 (2022), 102907.

Preprints and Manuscripts Under Review

[12] **C. Liu**, S. Tian, X. Liang, M. Zheng. "SELF-VLA: A Skill Enhanced Agentic Vision-Language-Action Framework for Contact-Rich Disassembly." *arXiv preprint* arXiv:2603.11080 (2026). Revised manuscript under review.

[13] S. Tian, **C. Liu**, M. Zheng, X. Liang. "NeurRAFT: Robot Motion Planning via Anchor-Level Flow Matching with Clearance-Aware Preference Tuning." *arXiv preprint* arXiv:2608.24026 (2026). Under review.

INDUSTRY EXPERIENCE

Senior Technical Support Engineer — Irvine, CA

HeyGears Inc. – Industrial 3D Printing — *Apr. 2021 – Nov. 2023*

- Managed technical support across an East Coast region serving 70+ clients and 300+ industrial 3D printers.
- Led new-product testing and customer troubleshooting; documented field issues to guide hardware and software improvements.

Mechanical Engineering Intern — Suzhou, China

Logitech International S.A. — *May 2019 – Aug. 2019*

- Consolidated mechanical engineering requirements and built a sample database and process instruction system for New Product Development Introduction (NPDI), using Axure and flowchart-based tooling.

Additional Research Projects

Perception and Planning for Disassembly (in progress) *Jan. 2024 – Present*

- Developing a pipeline combining YOLO detection, graph-based state representations, and vision-language reasoning for component selection and disassembly sequencing; preliminary results obtained.

Policy Generalization and Graph-Based World Models (in progress) *2025 – Present*

- Investigating adaptation under distribution shift using 3D scene reconstruction and human corrections; exploring scene-graph world models for disassembly-state prediction and VLA training data generation.

Precision 3D Profilometry for Remanufacturing *2024 – 2025*

- Contributed to high dynamic range fringe-projection scanning of computer enclosures for automated quality assessment (published in *Applied Optics*).

BSFL Monitoring and Growth Modeling *Jan. 2026 – Present*

- Built Raspberry Pi acquisition and monitoring for six experimental bins, with 30 continuous soil-temperature and moisture-voltage channels plus $O_2/CO_2/NH_3$ sampling.
- Implemented a sensor-only hierarchical GRU model for larval growth and harvest-window estimation; evaluated the prototype retrospectively with leave-one-bin-out validation on the six-bin pilot.

Erosion Function Apparatus (EFA) Automation *May 2025 – Present*

- Developing point-cloud measurement of soil erosion and automation for long-duration EFA tests with Dr. Jean-Louis Briaud's group in Civil and Environmental Engineering.

Mentoring & Professional Service

Graduate Mentor, Control and Robotics Lab, Summer 2026–Present. Mentoring two M.S. students in robot learning, experiment design, and technical writing; jointly preparing a journal submission.

Invited Reviewer, IEEE ICRA (2022, 2027), IEEE/RSJ IROS, and academic journals.

Awards & Honors

CoRL 2026 Travel Award, Conference on Robot Learning, 2026.

Graduate Student Travel Award, J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, FY 2026–2027; approved for CoRL 2026.

Graduate Summer Research Grant, J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, Summer 2026.

Technical Skills

Programming & Tools: Python, MATLAB/Simulink, SQL; PyTorch, TensorFlow, JAX, CUDA, Git, Docker, Linux, ZMQ
Robotics & Hardware: ROS2, MoveIt, Isaac Sim, UR5e, Teleoperation (GELLO, HOSAS), Hand-Eye Calibration, RealSense, OAK-D, Vicon, Raspberry Pi, SolidWorks
Perception & Policy Frameworks: YOLOv8, SAM2, Vision-Language Models, OpenVLA, OpenVLA-OFT, openpi
Languages: English (fluent), Mandarin Chinese (native)